

UrbanNest Home & Living

Online Store Process Improvement

A Business Analysis Case Study

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PROJECT TYPE: SIMULATED BUSINESS ANALYSIS CASE STUDY

UrbanNest Home & Living is a fictional company created for portfolio demonstration. All business data, metrics, stakeholder inputs and financial estimates are scenario assumptions. No real stakeholder interviews, ticket reviews or company engagement are claimed.

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****Version:**** 1.0

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1. Executive Summary

UrbanNest Home & Living, a mid-sized e-commerce retailer specializing in home décor and furniture, faced a critical challenge: despite steady website traffic, customer satisfaction was declining (NPS dropped from 45 to 32 over 18 months), and operational costs were rising.

This case study documents a comprehensive Business Analysis engagement to diagnose and redesign the company's customer order process. Through stakeholder interviews, process observation, data analysis, and root-cause investigation, I identified six interconnected problems: checkout friction, payment failures, inventory inaccuracy, order processing delays, poor status visibility, and return/refund friction.

The analysis revealed these problems were systemic—not isolated—and required a holistic process redesign rather than point fixes. I designed a future-state process that eliminates manual handoffs, enables real-time data flow, and provides proactive customer

communication.

****Expected outcomes:**** 25% reduction in cart abandonment, 40% faster order processing, 55% fewer status-related support tickets, and NPS improvement from 32 to 50+ within nine months.

2. Company & Business Context

Company Profile

Attribute	Details
Company Name	UrbanNest Home & Living
Founded	2016
Headquarters	Austin, Texas
Employees	85
Annual Revenue	\$12M (scenario estimate)
Product Categories	Furniture, home décor, kitchenware, bedding, lighting
Sales Channels	E-commerce website (85%), mobile app (15%)
Customer Base	45,000 active customers; ages 28-55; urban/suburban homeowners
Average Order Value	\$147
Fulfillment	2 warehouses (Austin, TX; Charlotte, NC); 3PL for oversized items

Operating Model

UrbanNest operates a hybrid fulfillment model:

- **Standard items**** (décor, kitchenware, bedding): Shipped from company warehouses
- **Oversized items**** (furniture, large lighting): Drop-shipped from 12 supplier partners
- **Customer Service**** In-house team of 8 agents
- **Technology Stack**** Shopify Plus, NetSuite ERP, custom middleware

Business Context

The company grew 40% year-over-year for three consecutive years. However, this growth exposed operational weaknesses. Customer complaints increased 55% in 12 months, with themes around order status confusion, delayed deliveries, and difficult returns. The CEO commissioned this initiative to address issues before peak season.

3. Business Problem

UrbanNest's order fulfillment process is fragmented across multiple systems and teams, resulting in:

- **High cart abandonment (68%)**** — Checkout friction and payment issues
- **Inventory discrepancies**** — Inaccurate stock levels causing overselling and cancellations
- **Order processing delays**** — Manual steps creating bottlenecks (9.2-day average delivery vs. 4-5 day benchmark)
- **Poor status visibility**** — Lack of accurate tracking generating 340 support tickets/month
- **Return/refund friction**** — Manual, slow, inconsistent return process (7-10 day refunds)

These problems are interconnected: inventory issues cause cancellations → refund requests → overwhelmed customer service → delayed response → reduced trust → increased cart abandonment.

Problem Statement

UrbanNest's customer order journey suffers from fragmented systems, manual handoffs, and poor communication, resulting in 68% cart abandonment, 12% payment failure, extended delivery times, and declining NPS (45 → 32).

4. Project Approach

Methodology

Phase	Activities	Duration
Discovery	Stakeholder interviews, process observation, data collection	Weeks 1-2
Analysis	AS-IS mapping, pain point identification, root-cause analysis	Weeks 3-5
Design	TO-BE process design, requirements documentation	Weeks 6-9
Validation	Stakeholder review, prioritization, business case	Weeks 10-12

Techniques Used

- Stakeholder interviews (10 stakeholders)
- Process observation (warehouse, customer service)
- Support ticket analysis (1,200 tickets reviewed)
- Data analysis (order timestamps, abandonment funnel)
- Fishbone root-cause analysis
- MoSCoW prioritization
- Process modeling (swimlane diagrams)

5. Current-State Analysis

AS-IS Process Overview

The current order process spans 11 stages across 6 systems with significant manual intervention.

Key Pain Points

ID	Pain Point	Severity	Frequency	Impact
PP-01	4-page checkout with mandatory account	Critical	Every order	High abandonment
PP-02	Payment failures with cryptic errors	Critical	12% of attempts	Lost sales
PP-03	No guest checkout option	High	Every new customer	Friction
PP-04	Inventory not real-time	Critical	23% of SKUs	Overselling
PP-05	2-hour batch sync delay	Medium	Every order	Processing delay
PP-06	Manual inventory verification	High	35% of orders	Bottleneck
PP-07	No order priority queue	Medium	Every order	Delayed expedited orders
PP-08	Tracking not synced to account	High	45% of orders	Support tickets
PP-09	No proactive delivery notifications	High	Every order	Customer anxiety
PP-10	24-48 hour CS response time	High	Peak periods	Dissatisfaction
PP-11	Manual return authorization	Medium	Every return	Delay
PP-12	7-10 day refund processing	High	Every refund	Top complaint

Pain Point Relationships

The pain points are not independent—they cascade:

6. Root-Cause Analysis

Fishbone Analysis

Category	Root Causes
People	Insufficient CS staffing; untrained staff; no exception ownership
Process	4-page checkout; no guest option; manual inventory; FIFO-only; manual returns
Technology	2-hour batch sync; no tracking integration; no notifications; gateway misconfigured
Data	Inventory not real-time; no centralized status; fragmented history
External	3PL limits; carrier API limits; payment processor rules

Root Cause Summary

Problem	Root Cause	Category
Cart abandonment	4-page checkout; mandatory account	Process
Payment failures	Gateway timeout misconfiguration	Technology
Inventory inaccuracy	2-hour batch sync	Technology/Data
Processing delays	Manual verification; no priority queue	Process
Poor visibility	Tracking not synced; no notifications	Technology
Return friction	Manual authorization; slow refunds	Process

7. Future-State Design

TO-BE Process Overview

The redesigned process eliminates manual handoffs, enables real-time data flow, and provides proactive communication.

Key Improvements

Area	AS-IS	TO-BE	Improvement
Checkout	4 pages, mandatory account	2 pages, guest option	50% fewer steps
Payment	12% failure, cryptic errors	≤4% failure, clear errors	67% reduction
Inventory	2-hour batch sync	5-minute real-time sync	96% faster
Order Routing	Manual, error-prone	Automated	100% elimination
Tracking	Not synced	Synced within 30 min	Real-time visibility
Notifications	None	Proactive at each stage	100% coverage
Returns	Manual approval	Self-service	Instant
Refunds	7-10 days	≤3 days	60% faster

Process Flow Comparison

****AS-IS:**** 16 steps, 6 manual handoffs, 4 systems

****TO-BE:**** 13 steps, 0 manual handoffs, integrated systems

8. Requirements & Prioritization

Requirements Summary

Category	Count	Must	Should	Could
Business Requirements	8	4	3	1
Functional Requirements	12	7	4	1
Non-Functional Requirements	8	5	3	0
User Stories	12	6	5	1

MoSCoW Prioritization

****Must Have (Phase 1-2):****

- Guest checkout (FR-01)
- 2-page checkout (FR-02)
- Real-time inventory (FR-03, FR-04)
- Payment error messages (FR-05)
- Auto order routing (FR-06)
- Tracking sync (FR-07)

****Should Have (Phase 3-4):****

- Proactive notifications (FR-08)
 - Self-service status (FR-09)
 - Online returns (FR-10)
 - 3-day refunds (FR-11)
- **Could Have (Phase 4):**
- Order prioritization (FR-12)

9. Expected Business Impact

Quantified Benefits (Scenario Estimates)

Benefit	Annual Value
Reduced cart abandonment	~\$324,000
Reduced payment failures	~\$144,000
Reduced support costs	~\$27,000
Reduced refund processing	~\$20,000
Increased repeat purchases	~\$120,000
Total	**~\$635,000**

KPI Targets

KPI	Baseline	Target	Timeline
Cart Abandonment	68%	≤50%	6 months
Payment Failure	12%	≤4%	3 months
Order-to-Delivery	9.2 days	≤6 days	6 months
Inventory Accuracy	87%	≥98%	4 months
Status Tickets	340/month	≤150/month	6 months
Refund Time	7-10 days	≤3 days	4 months
NPS	32	≥50	9 months

10. Recommendations

Phased Implementation

Phase	Timeline	Focus	Key Deliverables
1	Weeks 1-4	Checkout & Payment	Guest checkout; 2-page flow; payment optimization
2	Weeks 5-8	Inventory & Routing	Real-time sync; auto-routing
3	Weeks 9-12	Tracking & Notifications	Tracking integration; proactive notifications
4	Weeks 13-16	Returns & Refunds	Online returns; 3-day refunds

Critical Success Factors

1. Executive sponsorship and visible support
2. Cross-functional team collaboration
3. Rigorous testing before each phase launch
4. Clear customer communication during transitions
5. Data-driven monitoring of KPIs

11. Key BA Artifacts

Artifact	Format	Purpose
Stakeholder Matrix	Excel	Engagement planning
AS-IS Process Map	Diagram	Current state documentation
Pain Point Analysis	Excel	Problem identification
Fishbone Diagram	Diagram	Root cause analysis
TO-BE Process Map	Diagram	Future state design
Business Requirements	Excel	High-level needs
Functional Requirements	Excel	System capabilities
User Stories	Excel	Agile requirements
Acceptance Criteria	Excel	Testable conditions
MoSCoW Prioritization	Excel	Scope management
KPI Dashboard	Excel	Success measurement
Risk Matrix	Excel	Risk management
Implementation Roadmap	Diagram	Phased delivery

12. Skills Demonstrated

Skill Category	Specific Skills
Process Analysis	AS-IS mapping, TO-BE design, swimlane diagrams, process improvement
Root Cause Analysis	Fishbone diagrams, 5 Whys, problem relationship mapping
Requirements Elicitation	Stakeholder interviews, process observation, data analysis
Requirements Documentation	BRD, functional/non-functional requirements, user stories, acceptance criteria
Prioritization	MoSCoW method, effort/impact analysis, phased roadmap
Stakeholder Management	Stakeholder matrix, engagement strategies, communication planning
Data Analysis	KPI definition, baseline measurement, target setting
Risk Management	Risk identification, assessment, mitigation planning
Business Case	ROI estimation, benefit quantification, recommendation development
Tools & Techniques	Process modeling, requirements traceability, change management planning

13. Conclusion

UrbanNest's order process challenges are systemic and interconnected, requiring holistic redesign rather than isolated fixes. This analysis identified six core problems and traced them to root causes in process, technology, and data.

The recommended future-state process addresses each root cause through targeted improvements, with expected annual benefits of ~\$635K and NPS improvement from 32 to 50+.

Implementation should follow a phased approach, with quick wins in checkout and payment delivering early ROI. Success requires executive commitment, cross-functional collaboration, and rigorous change management.

This initiative positions UrbanNest to compete effectively on customer experience—a critical differentiator in the competitive home goods e-commerce market.
